Arrays
multidimensional arrays
example

Kees Huizing

Multi-dimensional arrays

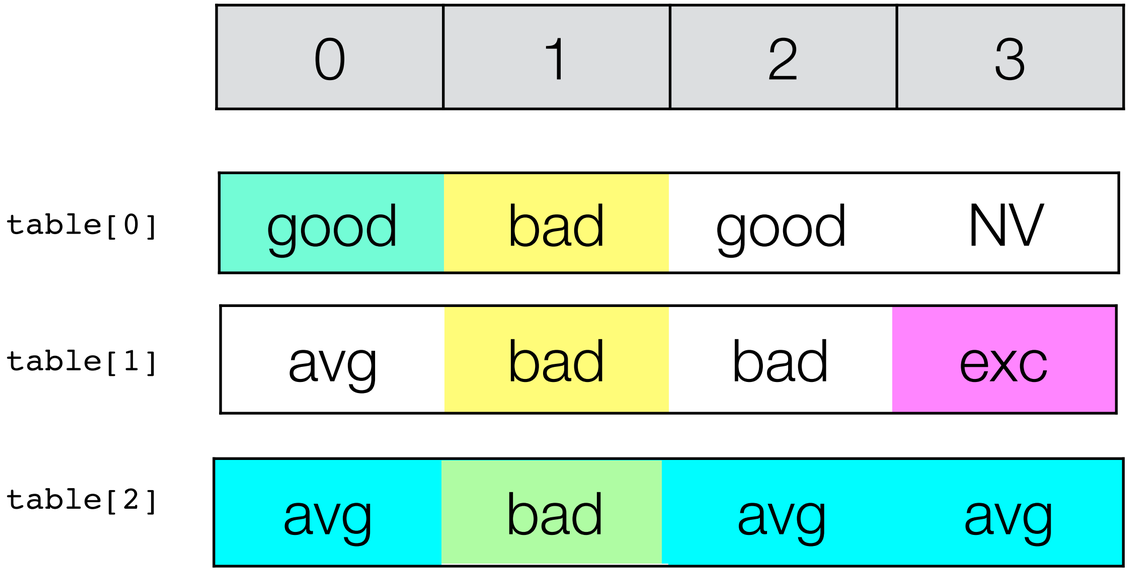
- to model grids, matrices, screens, tables, etc.
- example: student scores
  - student nr. *1* has scored an *exc* for course nr. *3*

course → student ↓	0	1	2	3
0	good	avg	good	NV
1	avg	avg	bad	exc
2	avg	avg	avg	avg

course → student ↓	0	1	2	3
0	good	bad	good	NV
1	avg	bad	bad	exc
2	avg		avg	avg

```
String[][] table;
```

in Java, a 2-dimensional array is an array of arrays



row/column layout

1st coordinate is vertical postion

```
String[][] table;

table = new String[3][4];

table[0][0] = "good";
table[1][3] = "exc";
for (int c=0; c < table[2].length; c++)
{
    table[2][c] = "avg";
}
for (int r=0; r < table.length; r++) {
    table[r][1] = "bad";
}
```

```
//print table
for (int r=0; r < table.length; r++) {
    for (int c=0; c < table[r].length; c++) {
        System.out.print( table[r][c] );
    }
    System.out.println();
}
```

in Java a 2-dimensional array is an array of arrays

table[0] table[1] table[2] table[3]

course →	0	1	2	3
0	good	bad	good	NV
1	avg	bad	bad	exc
2	avg	bad	avg	avg

x/y layout

1st coordinate is horizontal postion

```
String[][] table;

table = new String[4][3];

table[0][0] = "good";
table[3][1] = "exc";
for (int x=0; x < table.length; x++) {
    table[x][2] = "avg";
}
for (int y=0; y < table[1].length; r++) {
    table[1][y] = "bad";
}
//etc.
```

arbitrary index

```
//print table
for (int y=0; y < table[0].length; y++) {
     for (int x=0; x < table.length; x++) {
         System.out.print( table[x][y] );
     }
     System.out.println();
}
```

	table[0]	table[1]	table[2]	table[3]
course →	0	1	2	3
0	good	bad	good	NV
1	avg	bad	bad	exc
2	avg	bad	avg	avg

x/y layout

1st coordinate is horizontal postion

```
String[][] table;

table = new String[4][3];

table[0][0] = "good";
table[3][1] = "exc";
for (int x=0; x < table.length; x++) {
    table[x][2] = "avg";
}
for (int y=0; y < table[1].length; r++) {
    table[1][y] = "bad";
}
//etc.
```

arbitrary index

```
//print table
for (int y=0; y < table[0].length; y++) {
     for (int x=0; x < table.length; x++) {
         System.out.print( table[x][y] );
     }
     System.out.println();
}
```

Two-dimensional arrays

- conventions
  - pixels, geometry, etc.: first index is x-coordinate, second is y-coordinate
  - matrices: first index is row, second is column
  - different :-(
- it doesn’t matter which system you choose for your array, as long as you stick to your system

Ragged arrays

- what does this produce?

```
 0
10 11
20 21 22
30 31 32 33
40 41 42 43 44
```

```
//declaration
int[][] triangle;

//creation
triangle = new int[5][];
for (int r = 0; r < 5; r++) {
    triangle[r] = new int[r+1];
}

//filling
for (int r = 0; r < triangle.length; r++) {
    for (int c = 0; c < triangle[r].length; c++) {
        triangle[r][c] = 10*r + c;
    }
}

//printing
for (int r = 0; r < triangle.length; r++) {
    for (int c = 0; c < triangle[r].length; c++) {
        System.out.print(triangle[r][c] + " ");
    }
    System.out.println();
}
```

Frequency count (histogram)

- Suppose we want to count how often each grade (1-10) occurs in a list of 100 grades

```
int[] frequencies;
int grade;

frequencies = new int[11];

for (int i = 0; i<100; i++) {
    grade = scanner.nextInt();
    frequencies[ grade ] += 1;
}
```

```
for (int g = 0; g<frequencies.length; g++) {
    System.out.println("grade "+g+" occurs "+frequencies[g]+" time(s)");
}
```

Frequency count (histogram)

- Suppose we want to count how often each grade (1-10) occurs in a list of 100 grades

```
int[] frequencies;

frequencies = new int[11];
//assert all elts of frequencies are 0

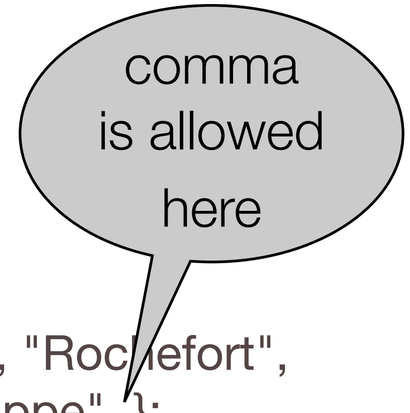
for (int i = 0; i<100; i++) {
    int grade = scanner.nextInt();
    assert 1<=grade && grade<=10;

    frequencies[ grade ] += 1;
}
```

More on arrays

- quick initialization:

```
String[ ] beers = {"Duvel", "Leffe Dubbel", "Hoare Tripel", };
```



- non-initialization:

```
trappists = new String[ ] { "Orval", "Chimay", "Westmalle", "Rochefort",
                            "Westvleteren", "Achel", "La Trappe", };

trappistWithScore = new String[ ][ ] {
                { "Orval", "great" }, { "Chimay", "great" }, { "Westmalle", "fine" },
                { "Rochefort", "excellent" }, { "Westvleteren", "excellent" },
                { "Achel", "fine" }, { "La Trappe", "good" },
            };
```

